

## BIRDSVILLE, Australia

WMO: 954820

Lat: 25.8975S

Lon: 139.3472E

Elev: 154

StdP: 14.61

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 40.6       | 42.8 | 7.2                           | 8.0 | 76.7 | 12.1 | 10.2 | 73.0 | 25.9                  | 70.2 | 21.9 | 68.9 | 5.6                   | 180  | 0.498 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 23.2                   | 110.2           | 69.8 | 107.9 | 69.3 | 105.6 | 68.7 | 79.0                | 91.5 | 76.9 | 90.6 | 75.2 | 90.4 | 11.2                 | 160  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 76.1                            | 137.2 | 83.8 | 73.4 | 124.9 | 83.4 | 70.6 | 113.5 | 83.5 | 42.7          | 91.4 | 40.5 | 90.4 | 38.8 | 90.3 | 99.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 26.7          | 22.8 | 20.3 | DB                         | 36.3 | 114.8              | 1.9 | 2.3                                                | 34.9 | 116.5      | 33.8 | 117.8      | 32.7 | 119.1      | 31.3 | 120.8 | (4) |
| (5)               |      |      | WB                         | 31.9 | 85.6               | 2.0 | 8.3                                                | 30.5 | 91.5       | 29.3 | 96.4       | 28.2 | 101.0      | 26.8 | 107.0 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 75.1  | 91.5  | 88.5  | 83.7  | 75.5  | 65.4  | 59.0  | 58.2  | 61.7  | 70.0  | 77.0  | 83.0  | 88.3  | (6)  |
| (7)  |                                                                                  | DBStd       | 13.24 | 6.02  | 5.80  | 6.41  | 6.38  | 6.06  | 5.73  | 5.31  | 6.57  | 7.43  | 7.48  | 7.08  | 6.53  | (7)  |
| (8)  |                                                                                  | HDD50       | 2     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 0     | 0     | 0     | 0     | 0     | (8)  |
| (9)  |                                                                                  | HDD65       | 678   | 0     | 0     | 0     | 4     | 70    | 200   | 223   | 146   | 31    | 4     | 0     | 0     | (9)  |
| (10) |                                                                                  | CDD50       | 9153  | 1287  | 1077  | 1045  | 765   | 477   | 270   | 255   | 364   | 601   | 837   | 989   | 1186  | (10) |
| (11) |                                                                                  | CDD65       | 4354  | 822   | 657   | 580   | 319   | 83    | 19    | 11    | 45    | 182   | 376   | 539   | 721   | (11) |
| (12) |                                                                                  | CDH74       | 64901 | 13609 | 10193 | 8143  | 3773  | 870   | 195   | 196   | 709   | 2512  | 5251  | 7936  | 11514 | (12) |
| (13) |                                                                                  | CDH80       | 40179 | 9465  | 6647  | 4837  | 1850  | 245   | 39    | 31    | 243   | 1233  | 2971  | 4906  | 7712  | (13) |
| (14) | Wind                                                                             | WSAvg       | 10.1  | 11.5  | 10.9  | 10.7  | 9.2   | 8.4   | 8.4   | 8.2   | 9.1   | 10.1  | 11.2  | 11.3  | 11.9  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 6.8   | 1.2   | 1.1   | 0.5   | 0.4   | 0.6   | 0.4   | 0.5   | 0.4   | 0.2   | 0.5   | 0.5   | 0.7   | (15) |
| (16) |                                                                                  | PrecMax     | 19.5  | 10.4  | 5.0   | 5.2   | 3.4   | 3.9   | 1.9   | 4.3   | 2.2   | 0.9   | 2.6   | 1.9   | 4.0   | (16) |
| (17) |                                                                                  | PrecMin     | 1.3   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | (17) |
| (18) |                                                                                  | PrecStd     | 3.5   | 2.0   | 1.5   | 1.0   | 0.8   | 0.8   | 0.5   | 0.8   | 0.5   | 0.2   | 0.6   | 0.5   | 0.9   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 113.5 | 110.8 | 107.9 | 99.4  | 89.5  | 84.9  | 83.9  | 92.8  | 101.8 | 106.0 | 110.9 | 112.9 | (19) |
| (20) |                                                                                  |             | MCWB  | 70.9  | 70.9  | 69.5  | 65.8  | 62.2  | 63.0  | 57.7  | 62.3  | 62.9  | 65.7  | 68.5  | 68.7  | (20) |
| (21) |                                                                                  | 2%          | DB    | 110.6 | 107.6 | 103.7 | 96.0  | 85.8  | 80.3  | 79.2  | 86.4  | 96.4  | 101.9 | 106.7 | 109.3 | (21) |
| (22) |                                                                                  |             | MCWB  | 71.0  | 69.9  | 69.0  | 65.1  | 61.3  | 60.2  | 56.8  | 59.3  | 62.3  | 65.0  | 66.7  | 68.2  | (22) |
| (23) |                                                                                  | 5%          | DB    | 107.9 | 104.9 | 100.5 | 93.2  | 82.7  | 75.3  | 75.2  | 81.8  | 91.9  | 98.5  | 102.9 | 106.3 | (23) |
| (24) |                                                                                  |             | MCWB  | 70.6  | 69.7  | 68.4  | 63.9  | 60.3  | 57.8  | 55.5  | 57.0  | 60.5  | 63.6  | 66.0  | 68.1  | (24) |
| (25) |                                                                                  | 10%         | DB    | 105.3 | 102.0 | 97.9  | 90.0  | 79.1  | 71.2  | 71.3  | 77.3  | 87.6  | 94.7  | 99.2  | 103.6 | (25) |
| (26) |                                                                                  |             | MCWB  | 69.7  | 69.6  | 67.5  | 63.1  | 58.9  | 55.5  | 53.9  | 55.1  | 58.9  | 62.5  | 65.2  | 67.7  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 80.8  | 81.3  | 78.5  | 74.7  | 68.7  | 65.4  | 64.8  | 65.1  | 70.2  | 73.4  | 78.2  | 88.4  | (27) |
| (28) |                                                                                  |             | MCD B | 91.5  | 90.4  | 91.8  | 84.0  | 78.0  | 79.3  | 72.7  | 85.0  | 84.6  | 86.7  | 92.9  | 96.0  | (28) |
| (29) |                                                                                  | 2%          | WB    | 78.0  | 78.2  | 76.1  | 71.4  | 65.3  | 62.7  | 60.6  | 61.8  | 66.1  | 69.2  | 73.0  | 77.4  | (29) |
| (30) |                                                                                  |             | MCD B | 92.0  | 91.5  | 87.3  | 82.2  | 77.7  | 74.5  | 71.6  | 79.1  | 82.8  | 88.2  | 87.5  | 89.9  | (30) |
| (31) |                                                                                  | 5%          | WB    | 76.0  | 76.5  | 74.2  | 68.3  | 63.1  | 60.2  | 57.7  | 59.0  | 63.6  | 66.6  | 70.7  | 74.0  | (31) |
| (32) |                                                                                  |             | MCD B | 92.3  | 90.9  | 88.2  | 84.7  | 77.0  | 71.1  | 71.4  | 77.4  | 84.6  | 89.0  | 88.9  | 90.9  | (32) |
| (33) |                                                                                  | 10%         | WB    | 74.2  | 74.9  | 72.0  | 66.0  | 61.0  | 57.7  | 55.2  | 56.4  | 61.1  | 64.7  | 69.0  | 72.0  | (33) |
| (34) |                                                                                  |             | MCD B | 93.4  | 91.6  | 88.7  | 84.4  | 74.8  | 68.6  | 68.9  | 74.9  | 83.2  | 89.2  | 89.1  | 92.1  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 23.2  | 22.5  | 23.0  | 24.4  | 23.4  | 22.4  | 24.5  | 26.8  | 27.7  | 27.2  | 25.8  | 24.8  | (35) |
| (36) |                                                                                  |             | MCDBR | 25.1  | 26.1  | 26.1  | 27.2  | 26.4  | 26.7  | 29.6  | 32.0  | 33.6  | 31.7  | 30.4  | 28.5  | (36) |
| (37) |                                                                                  |             | MCWBR | 9.4   | 10.3  | 10.5  | 11.3  | 12.1  | 13.1  | 14.9  | 15.4  | 15.2  | 14.0  | 12.4  | 11.4  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 20.3  | 18.8  | 19.5  | 21.3  | 22.2  | 21.2  | 24.4  | 27.0  | 26.6  | 25.4  | 23.6  | 22.2  | (38) |
| (39) |                                                                                  |             | MCWBR | 9.7   | 8.6   | 9.4   | 9.6   | 11.1  | 11.4  | 13.5  | 14.0  | 13.5  | 14.1  | 11.7  | 12.5  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.387 | 0.381 | 0.359 | 0.337 | 0.319 | 0.306 | 0.306 | 0.325 | 0.374 | 0.378 | 0.388 | 0.390 | (40)  |      |
| (41) |                                                                                  | taud        | 2.348 | 2.375 | 2.406 | 2.430 | 2.442 | 2.465 | 2.445 | 2.372 | 2.249 | 2.258 | 2.288 | 2.311 | (41)  |      |
| (42) |                                                                                  | Ebn at Noon | 303   | 300   | 298   | 289   | 281   | 279   | 283   | 289   | 288   | 298   | 301   | 303   | (42)  |      |
| (43) |                                                                                  | Edh at Noon | 43    | 41    | 37    | 33    | 30    | 28    | 29    | 35    | 43    | 45    | 45    | 44    | (43)  |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2515  | 2378  | 2103  | 1759  | 1388  | 1213  | 1323  | 1642  | 1980  | 2316  | 2446  | 2498  | (44)  |      |
| (45) |                                                                                  | RadStd      | 123   | 140   | 125   | 98    | 78    | 74    | 66    | 73    | 118   | 119   | 125   | 138   | (45)  |      |

## Historical Trends

|      |                        | Heating |        | Cooling |       |       | Degree-Days |       |       |       |     |
|------|------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|-----|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |     |
|      |                        | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m) |
| (46) | Station Only           | N/A     | N/A    | N/A     | +2.21 | N/A   | N/A         | N/A   | N/A   | N/A   | N/A |
| (47) | Regional (0 neighbors) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   | N/A |

Nomenclature: See separate page